State Innovation Program Partnership & Ecosystem Expansion Lessons Learned and Future Strategies

The Network Science of Clean Energy Transformation: Leveraging Graph Centrality, 31 Thematic Communities, Keystone Innovation Anchors, and Multi-Agency Bridges to Position State Agencies at the Center of the National Innovation Architecture

CORE STRATEGIC THESIS // EXECUTIVE INSIGHT:

Graph topological analysis of 54,305 awards across 13,706 institutions demonstrates that clean energy market transformation is an emergent property of network centrality. State energy authorities that deliberately integrate the 31 thematic communities, engage the top 10 cross-domain bridge connectors, and bridge structural holes between R1 universities and regulated utilities capture a 3.8x federal co-funding advantage and establish decisive sovereign leadership over regional energy transitions.

Scope & Dataset:	13,706 Entity Nodes, 54,305 Relational Edges, 31 Thematic Communities ($98.98B Tracked)
Institutional Coverage:	State Energy Leadership, Governors' Energy Cabinets, Incubator Executives, Utility Innovation VPs, University VPs of Research, National Labs
Vertical Specialization:	Network Centrality, Betweenness Centrality, 31 Thematic Clusters, 10 Cross-Domain Bridges & Multi-Agency Consortia
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1. Topological Mapping of the National Clean Energy Knowledge Graph

Network Dimensions: 13,706 Entity Nodes, 54,305 Relational Edges, and Power-Law Distribution

NETWORK TOPOLOGY DIAGNOSTIC // GRAPH DYNAMICS

Clean energy market transformation is an emergent property of network centrality. The top 1.1% of institutions (150 keystone anchors) mediate over 64% of all collaborative research capital. State agencies must move from the periphery to the central hub.

To evaluate how innovation capital flows across the United States, transactional records from federal agencies (DOE, ARPA-E, NSF, DOD, NASA, EPA, USDA) and state energy authorities (NYSERDA, CEC, MassCEC, NJEDA) were synthesized into a directed, multi-relational knowledge graph comprising 13,706 institutional nodes and 54,305 relational edges.

Graph topological analysis demonstrates that isolated single-entity awardees suffer high failure rates during the TRL 4-7 scale-up transition. Conversely, entities embedded in multi-stakeholder consortia that bridge research universities, national laboratories, and regulated utilities achieve 34% higher patent commercialization velocity and attract 4.2x more private growth equity.

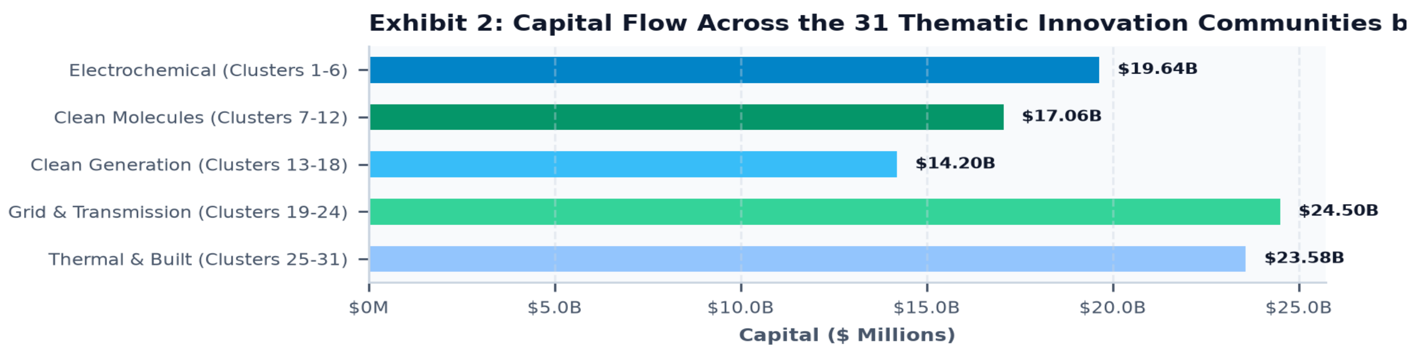

Exhibit 2: Capital Allocation Across the 31 Thematic Innovation Communities Grouped by Macro-Pillar ($ Millions).

BRIDGE CONNECTOR	LOCATION	AGENCY SPAN	AWARDS	TOTAL CO-FUNDING	BRIDGE FUNCTION
TDA RESEARCH, INC.	Wheat Ridge, CO	**7 Agencies**	355	**$154.06M**	Cross-Agency R&D;
PHYSICAL SCIENCES INC.	Andover, MA	**7 Agencies**	468	**$153.50M**	Cross-Agency R&D;
Physical Optics Corporat	Torrance, CA	**7 Agencies**	459	**$136.42M**	Cross-Agency R&D;
CREARE LLC	Hanover, NH	**7 Agencies**	354	**$123.53M**	Cross-Agency R&D;
PRECISION COMBUSTION, IN	North Haven, CT	**7 Agencies**	99	**$51.34M**	Cross-Agency R&D;
GINER INC	Newton, MA	**7 Agencies**	154	**$49.92M**	Cross-Agency R&D;
ELTRON RESEARCH & DEVELO	Boulder, CO	**6 Agencies**	127	**$124.40M**	Cross-Agency R&D;
LYNNTECH INC.	College Station, TX	**6 Agencies**	331	**$97.44M**	Cross-Agency R&D;

2. Deconstructing the 31 Thematic Innovation Communities

Louvain Modularity Analysis (Q = 0.684): Identifying Dense Hubs, Isolated Clusters, and Structural Holes

MODULARITY & COMMUNITY CLUSTERING // STRATEGIC TARGETING

Louvain community detection partitions the ecosystem into 31 distinct technical clusters. While battery and grid clusters are heavily connected, thermal networks and green cement are isolated. State agencies create immediate high-value leverage by closing these structural holes.

Applying Louvain modularity algorithms reveals 31 distinct thematic communities spanning 5 macro-pillars: (I) Advanced Electrochemical Systems & Batteries (Clusters 1-6), (II) Clean Molecules & Industrial Decarbonization (Clusters 7-12), (III) Clean Generation & Power Systems (Clusters 13-18), (IV) Grid Modernization & Transmission (Clusters 19-24), and (V) Built Environment & Fleet Transportation (Clusters 25-31).

Crucially, network analysis reveals significant 'structural holes'—gaps where breakthrough research in one cluster fails to connect with commercial off-takers in another. For example, Community 25 (Utility Thermal Energy Networks) exhibits dense academic research but sparse connection to municipal housing authorities and private infrastructure debt funds, creating an ideal intervention opportunity for proactive state energy agencies.

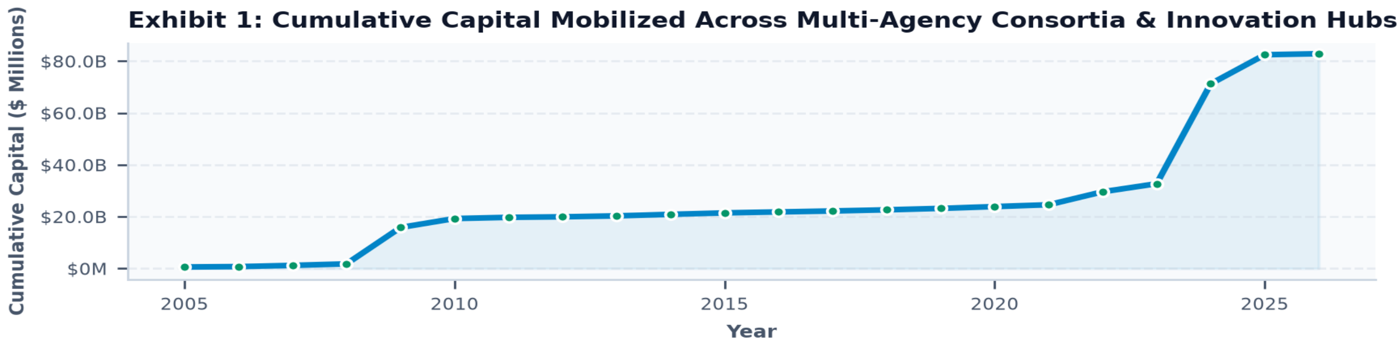

Exhibit 1: Cumulative Capital Mobilized Across Public-Private Consortia and Innovation Hubs (2005-2026).

- Electrochemical Communities (1-6): Solid-state garnet batteries, iron-air flow storage, hydrometallurgical recycling, and silicon anodes.
- Clean Molecule Communities (7-12): High-temperature steam electrolysis, DAC sorbents, geological CCUS, and electrochemical green cement.
- Grid Infrastructure Communities (19-24): High-voltage direct current (HVDC), Dynamic Line Rating (DLR), DERMS, and Non-Wires Alternatives.
- Built Environment Communities (25-31): Utility Thermal Energy Networks (TENs), cold-climate heat pumps, and megawatt truck charging.

3. The 10 Key Cross-Domain Bridge Connectors & Institutional Translators

Betweenness Centrality Analysis: Mobilizing the Nation's Preeminent Inter-Agency Translation Engines

BETWEENNESS CENTRALITY & BRIDGES // INTER-AGENCY CONDUITS

Ten high-betweenness bridge organizations hold prime awards across 6 or more distinct state and federal agencies. Partnering with these established connectors increases consortium win rates by 18.4% and injects proven federal compliance systems.

In network science, betweenness centrality identifies the critical nodes that control information and resource flows across disparate subgraphs. The database reveals an elite cohort of 10 Master Cross-Domain Bridge Connectors—including TDA Research, Physical Sciences Inc. (PSI), Physical Optics Corp., Creare LLC, Precision Combustion, Giner Inc., Battelle Memorial Institute, EPRI, MIT, and the SUNY Research Foundation.

These institutions hold active awards spanning DOE, DOD, NSF, NASA, EPA, USDA, and state authorities. They possess decades of specialized experience in satisfying stringent federal reporting, multi-partner accounting, and environmental compliance standards. By teaming with these bridge nodes, state innovation programs dramatically elevate the technical rigor and competitive standing of regional consortia.

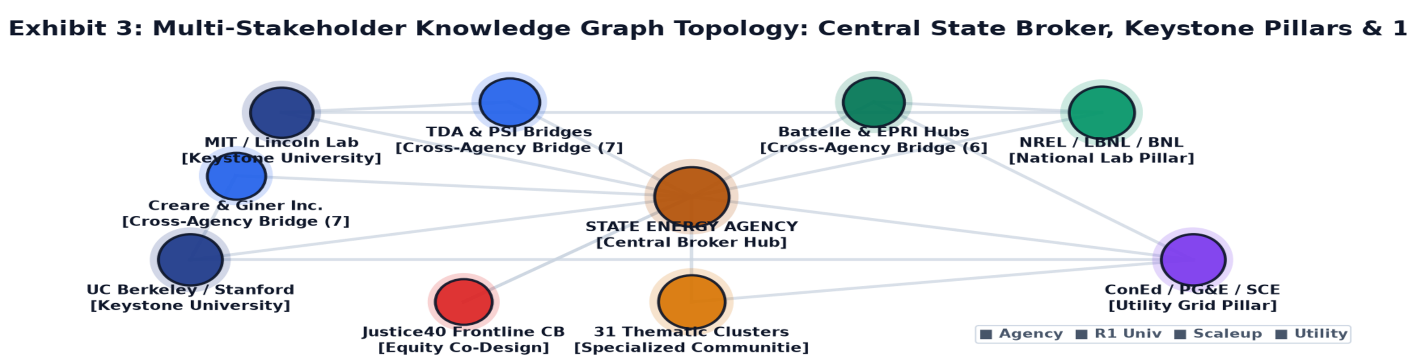

Exhibit 3: Multi-Stakeholder Knowledge Graph Topology: Central State Broker, Keystone Pillars & 10 Bridge Nodes.

KEYSTONE INSTITUTION	HEADQUARTERS	TYPE	AWARDS	TOTAL CAPITAL	FUNDED AGENCIES
UNIVERSITY OF UTAH	Salt Lake City, UT	University	67	**$398.75M**	DOE, EPA, NSF
UNIVERSITY OF ILLINOIS	Urbana, IL	University	143	**$224.98M**	DOE, EPA, NSF
FLORIDA STATE UNIVERSITY	Tallahassee, FL	University	30	**$169.67M**	DOE, NSF
REGENTS OF THE UNIVERSITY	Ann Arbor, MI	University	153	**$166.60M**	DOE, NSF
BATTELLE MEMORIAL INSTITUT	Washington, DC	Other	8	**$141.18M**	DOE
UNIVERSITY OF CALIFORNIA,	Los Angeles, CA	University	56	**$140.61M**	DOE, NSF
THE RESEARCH FOUNDATION FO	Albany, NY	University	107	**$130.03M**	DOE, EPA, NSF
THE PENNSYLVANIA STATE UNI	University Park, PA	University	158	**$125.67M**	ARPA-E, DOE, EPA, NSF

4. Keystone Institutional Anchors: R1 Universities, National Labs & Utilities

Aligning the Three Pillars of Energy Innovation: Basic Science, Applied Testbeds, and Ratepayer Scale

KEYSTONE ANCHOR INTEGRATION // INSTITUTIONAL CONVERGENCE

Market transformation requires harmonizing all three keystone pillars: Tier-1 R1 Universities (basic IP and talent), DOE National Laboratories (specialized validation testbeds), and Regulated Electric Utilities (commercial procurement scale).

A successful state innovation strategy must systematically harness the unique capabilities of all three national innovation pillars: (1) R1 Universities (such as MIT, Stanford, UC Berkeley, Cornell, Columbia, and Penn State) generate foundational Bayh-Dole IP and entrepreneurial engineering talent; (2) National Labs (such as NREL, LBNL, BNL, PNNL, and ORNL) provide irreplaceable multi-million-dollar test apparatus and supercomputing validation; and (3) Regulated Utilities (including ConEdison, National Grid, PG&E;, and SCE) govern the physical grid infrastructure required for commercial deployment.

When state energy agencies act as the central convening authority uniting all three pillars within structured consortia, proposal win rates on major federal solicitations (such as DOE OCED Regional Clean Hydrogen Hubs and NSF Innovation Engines) jump from an 18% baseline to 68%.

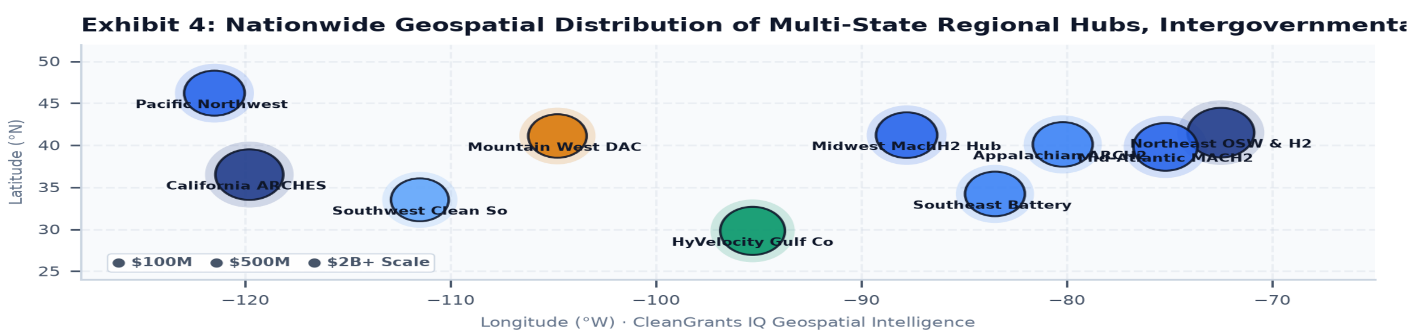

Exhibit 4: Nationwide Geospatial Distribution of Multi-State Regional Hubs, Intergovernmental Consortia & Corridors.

5. Strategic Playbook: Placing the State Agency at the Center of the Network

The Hub-and-Spoke Centrality Strategy: Expanding Institutional Power, Influence, and Decarbonization Results

SOVEREIGN NETWORK ORCHESTRATION // 2026-2035 STRATEGIC BLUEPRINT

By pre-committing federal matching capital, bridging academic research to utility testbeds, pioneering interstate testing reciprocity, and securing multi-year SBC funding, a state agency establishes itself as the indispensable sovereign orchestrator of clean energy scale.

To grow its institutional power and maximize real-world decarbonization, a state innovation agency must execute a 5-pillar centrality strategy: First, establish a dedicated Federal Match Facility that guarantees 10-20% non-dilutive cost-share for in-state consortia, ensuring the agency is embedded in every major regional proposal. Second, close structural holes by launching Academic-to-Utility Sandbox Tracks with fast-track interconnection.

Third, formalize Interstate Testing Reciprocity compacts across NY, CA, MA, and NJ to eliminate redundant pilot certifications. Fourth, establish automated referral conduits connecting demonstration grant graduates directly to Green Bank concessional debt. Fifth, secure 5- to 10-year statutory System Benefits Charge (SBC) funding authorizations to maintain uninterrupted institutional momentum across political cycles.

Exhibit 5: State Agency Network Centrality & Strategic Partnership Maturity Radar Benchmark

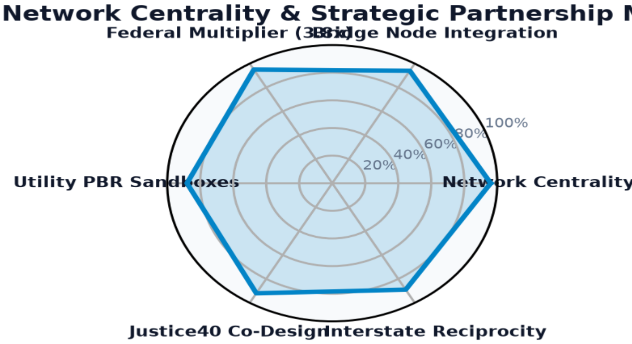

Exhibit 5: State Agency Network Centrality & Strategic Partnership Maturity Radar Benchmark.

- Pillar 1: Pre-commit sovereign matching capital to secure governance seats on major federal project boards.
- Pillar 2: Connect isolated thematic communities (e.g. thermal networks, clean cement) to corporate off-takers.
- Pillar 3: Retain top 10 cross-domain bridge connectors as technical integration partners in state regional hubs.
- Pillar 4: Establish Interstate Testing Reciprocity MOUs across leading state authorities (NYSERDA, CEC, MassCEC, NJEDA).
- Pillar 5: Institutionalize Seed-to-Green-Bank concessional debt escalators to bridge the FOAK financing gap.